

Mohammad Sadra Moazzen

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OBJECTIVE

Aspiring Computational Neuroscientist and NeuroAI Researcher with expertise in cell and molecular biology, data science, and artificial intelligence. Passionate about applying advanced analytical techniques and machine learning to unravel complex biological questions, particularly in computational neuroscience and neuroimaging.

EDUCATION

Master of Science in Cell and Molecular Biology Hakim Sabzevari University	2025 - 2027
MicroMaster in Artificial Intelligence Sharif University of Technology	2023 - 2026 GPA: 4.0 / 4.0
MicroMaster in Data Science Sharif University of Technology	2023 - 2024 GPA: 4.0 / 4.0
Bachelor of Science in Cell and Molecular Biology Ferdowsi University of Mashhad	2021 - 2025 GPA: 3.64 / 4.0

SKILLS

Programming Languages	Python, MATLAB, R, C/C++
Data Science Tools	SQL, NumPy, Pandas, Matplotlib, Seaborn, Bokeh
Machine Learning Frameworks	TensorFlow, PyTorch, Scikit-learn, OpenCV
Neuroimaging Tools	SPM, CONN, FREESURFER, FSL, Nilearn
Computational Neuroscience Tools	NWB, NeuroDSP, MNE, NetworkX, Elephant

INTERESTS

Computational Neuroscience, NeuroAI, Machine Learning

RESEARCH EXPERIENCE

MIT (CompNeuro) <i>Researcher (Remote)</i>	August 2024 - February 2025
- Working with Dr. Nima Dehghani on one research paper: <i>Decoding Mouse Sleep Stage Neural Dynamics with Time Delay Stability, Phase Synchronization, Neural Granger Causality, and Auxiliary-Driven Self-Supervised Embedding</i>	
Research Institute of Biotechnology <i>Undergraduate Researcher (In-Person)</i>	July 2023 - August 2026
- Worked with Dr. Fatemeh B. Rassouli on two research papers: <i>Maximizing Efficacy: Machine Learning-Driven Dose-Time Predictions for Coumarin Therapies.</i> <i>In Vitro Insights: Machine Learning-Driven Evaluation of Esculetin's Anticancer Action.</i>	
Interdisciplinary Schools <i>Researcher (Remote)</i>	September 2023 - November 2025
- Worked with Dr. Abolhasan Rezaeyan on one research paper: <i>Temporal Lobe Epilepsy Connectivity: Insights from MRI and Machine Learning.</i>	

PUBLICATIONS

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Machine learning-guided dose-time optimization and experimental validation enhance coumarin therapeutics in oncology

Scientific Reports Published

- Used machine learning to identify optimal dose-time combinations for enhancing the anticancer efficacy of natural coumarins.

Pooled Evidence Supports Esculetin as a Promising Natural Coumarin for Cancer Therapy

Journal of Pharmaceutical Innovation Published

- Conducted a systematic evaluation of in vitro evidence on the anticancer effects of esculetin across human cancers.

Temporal Lobe Epilepsy Connectivity: Insights from MRI and Machine Learning Under Review

- Applied machine learning to classify patients with temporal lobe epilepsy using structural and functional connectivity features derived from MRI and resting-state fMRI.

PROJECTS

Enhancing Deep Learning Robustness in Microscopy Images Project

Programmer July 2024

- Improved U-Net’s robustness in microscopy image segmentation by addressing noise and artifacts.
- Tools: Python, PyTorch, OpenCV, Matplotlib.
- Project repository: GitHub

Sternberg Memory Task

Programmer March 2024

- Simulated a classic psychological experiment on short-term memory using Python.
- Tools: Python, Pygame, Matplotlib.
- Project repository: GitHub

TEACHING EXPERIENCE

Neuroscience and AI

Shahid Beheshti University of Medical Sciences November 2025

- Delivered six in-depth sessions on the intersection of neuroscience and artificial intelligence, highlighting core neural mechanisms, recent research advances, and their applications in cognitive modeling and AI systems.

Machine Learning in Neuroimaging

Interdisciplinary Schools February 2025 - June 2025

- Designed and taught a course on ML applications in neuroimaging.

Machine Learning Techniques

Interdisciplinary Schools December 2024 - February 2025

- Taught a course on Machine Learning techniques and their applications.

Data Science

Interdisciplinary Schools July 2024 - October 2024

- Taught a Data Science course covering NumPy, Pandas, Matplotlib, and Seaborn.

TEACHING ASSISTANT POSITIONS

Computational Neuroscience
NeuroCONNECT

August 2026

- Led a group of 15 students over 3 weeks in hands-on computational neuroscience education. Guided interactive online classrooms and mentored research projects. Topics covered included spiking neuron models, synaptic plasticity, neural dynamics, reinforcement learning, probabilistic models, and deep learning in neuroscience. Provided expert support in computational modeling and neuroscience methodologies, fostering a collaborative and inclusive learning environment.

Computational Neuroscience
The Neuromatch Academy

July 2025

- Led a group of 14 students over 3 weeks in hands-on computational neuroscience education. Guided interactive online classrooms and mentored research projects. Topics covered included spiking neuron models, synaptic plasticity, neural dynamics, reinforcement learning, probabilistic models, and deep learning in neuroscience. Provided expert support in computational modeling and neuroscience methodologies, fostering a collaborative and inclusive learning environment.

Python for Non-Engineers
Interdisciplinary Schools

December 2024 - February 2025

- Supported the course by introducing Python fundamentals, facilitating hands-on exercises, guiding discussions, and ensuring accessibility for participants from diverse academic backgrounds.

VOLUNTEER POSITIONS

Course Coordinator and Content Lead: Python in Neuroimaging (fMRI)
Interdisciplinary Schools

July–October 2025

- Oversaw lesson design and delivery quality for an fMRI-focused Python module by coordinating with the instructor and assistant through regular meetings, ensuring pedagogical coherence, technical accuracy, and alignment with best practices in neuroimaging education.

Collaborator: Software and Programming Languages
Interdisciplinary Schools

October 2024 - July 2026

- Contributed to software development and optimization projects, utilizing expertise in programming languages and frameworks to deliver scalable, high-quality solutions.

Course Developer & Instructor: Computational Neuroscience
Interdisciplinary Schools

September 2024 - July 2026

- Developed and instructed a course in computational neuroscience, focusing on the design of comprehensive educational materials.

Mentor: Neuroimaging Group
Interdisciplinary Schools

July 2024 - August 2026

- Mentored a four-person neuroimaging team, providing guidance to enhance learning, support research initiatives, and foster innovative problem solving.

Editor-in-Chief & Writer
Ferdowsi University of Mashhad

December 2022 - December 2023

- Volunteered as Editor-in-Chief of the Biology Scientific Association, leading content creation and editorial processes.

CERTIFICATIONS

Deep Learning <i>Neuromatch Academy</i>	July 2024
Machine Learning Specialization <i>DeepLearning.AI</i>	April 2024
Understanding the Brain: The Neurobiology of Everyday Life <i>University of Chicago</i>	August 2023
CS50's Introduction to Programming with Python <i>CS50</i>	August 2023
Data Analyst Professional Certificate <i>IBM</i>	October 2023
CS50's Introduction to Databases with SQL <i>CS50</i>	November 2023
Cognitive Neuroscience <i>Shahid Beheshti University</i>	November 2023
Computational Neuroscience <i>Neuromatch Academy</i>	July 2023